

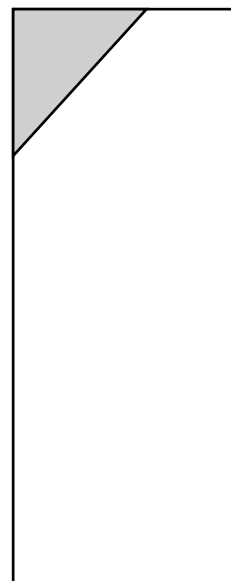
Set 4 - Challenge Practice

Level: Hard

Name: _____ Class: _____ Date: _____

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1. A rectangular paper measures 48 cm by 22 cm. Two corner triangles are folded in. Triangle Q has area 36 cm^2 and Triangle P has twice the area of Triangle Q. Find the shaded area left.

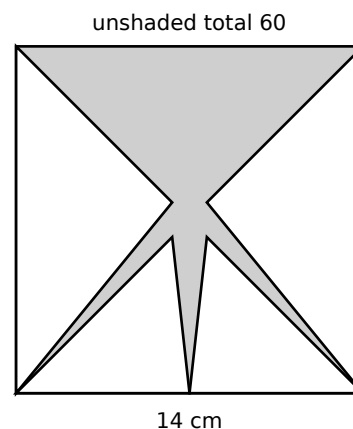


rectangle 48×22 ; $P = 2Q$; $Q = 36$

Working:

Answer:

2. A square PQRS has side 14 cm. The total area of four unshaded corner triangles is 60 cm^2 . Find the shaded area.



Working:

Answer:

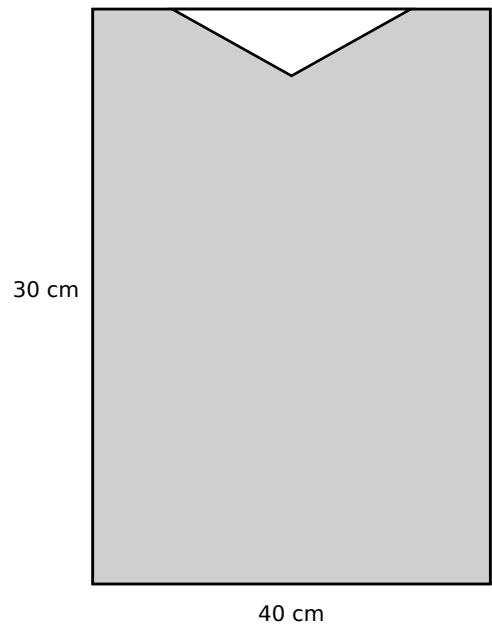
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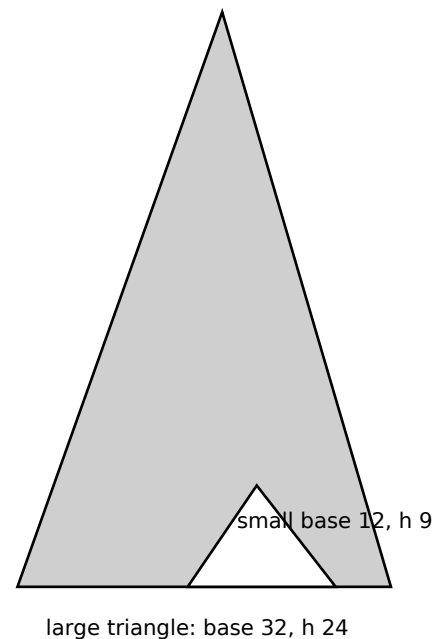
- 3.** A rectangle measures 40 cm by 30 cm. A triangular part with base 18 cm and height 20 cm is unshaded. Find the shaded area.



Working:

Answer:

- 4.** A large right-angled triangle has base 32 cm and height 24 cm. A smaller unshaded triangle has base 12 cm and height 9 cm. Find the shaded area.



Working:

Answer: _____

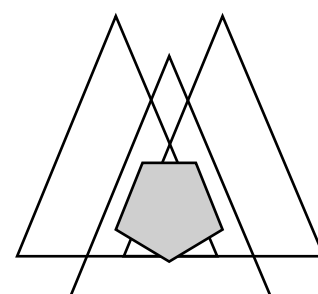
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5. A pattern is made of 5 identical triangles. Each triangle has base 12 cm and height 10 cm. Three triangles are shaded. Find the shaded area.

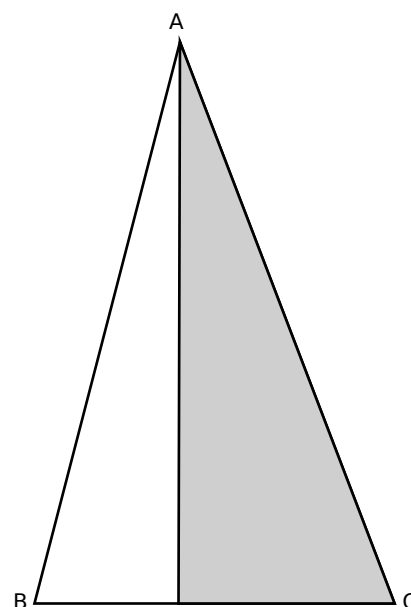


5 identical triangles; 3 shaded

Working:

Answer:

6. In Triangle ABC, BC is divided at D in the ratio $BD:DC = 2:3$. The area of Triangle ABC is 240 cm^2 . Find the area of Triangle ADC.



BC split 2:3
same height

Working:

Answer:

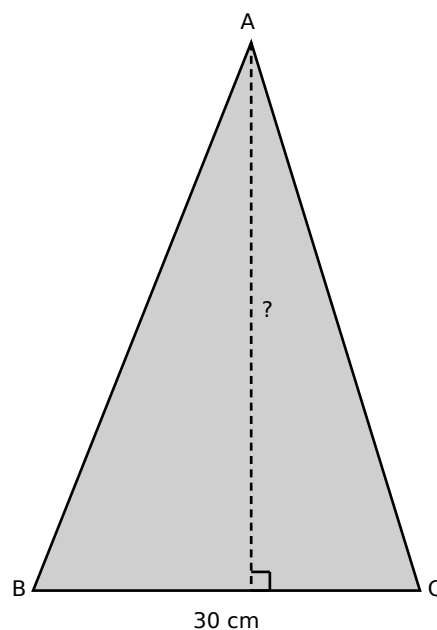
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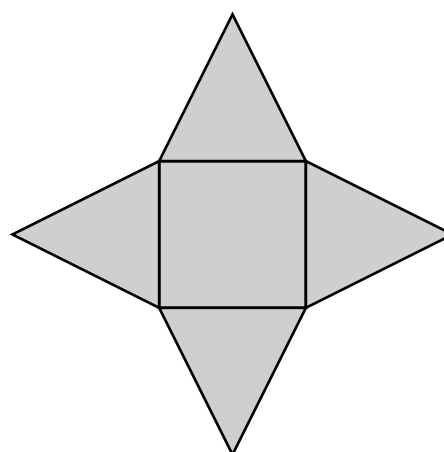
7. Triangle AEB has base $AB = 30\text{ cm}$ and area 135 cm^2 . Find the perpendicular height from E to AB.



Working:

Answer:

8. A figure is made up of a square and 4 identical triangles. The square has side 10 cm. Each triangle has base 10 cm and height 8 cm. Find the total area.



square side 10 + 4 triangles

Working:

Answer:

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- 9.** Lina donated $\frac{3}{8}$ of her books and sold $\frac{2}{5}$ of the remainder. She had 45 books left. How many books did she have at first?

Working:

Answer:

- 10.** A tank had some water. $\frac{1}{3}$ was used first, then $\frac{1}{4}$ of the remainder, then $\frac{2}{5}$ of the new remainder. If 24 litres were left, how much water was in the tank at first?

Working:

Answer:

Set 4 - Answer Sheet

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Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	948 cm ²	Rectangle = 1056. Q = 36, P = 72. Shaded = 1056 - 108 = 948 cm ²
2	136 cm ²	Square = 196; shaded = 196 - 60 = 136 cm ²
3	1020 cm ²	Rectangle = 1200; triangle = $\frac{1}{2} \times 18 \times 20 = 180$; shaded = 1020 cm ²
4	330 cm ²	Large = 384; small = 54; shaded = 330 cm ²
5	180 cm ²	Each triangle = $\frac{1}{2} \times 12 \times 10 = 60$; 3 shaded = 180 cm ²
6	144 cm ²	DC is $\frac{3}{5}$ of BC. Area ADC = $\frac{3}{5} \times 240 = 144$ cm ²
7	9 cm	$135 = \frac{1}{2} \times 30 \times h$, so $h = 270 / 30 = 9$ cm
8	260 cm ²	Square = 100; each triangle = 40; 4 triangles = 160; total = 260 cm ²
9	120 books	Left = $\frac{5}{8} \times \frac{3}{5} = \frac{3}{8}$ of total. $\frac{3}{8} = 45$, total = 120
10	80 litres	Left = $\frac{2}{3} \times \frac{3}{4} \times \frac{3}{5} = \frac{3}{10}$ of total. $\frac{3}{10} = 24$, total = 80